

Cluster Access and SING Quantization

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Updates

- Got access to sassauna servers containing NVIDIA GeForce GTX 1080 GPUs
- Dealt with library and Cuda version issues
- Tried to replicate results from SING paper table 1
- Attempted quantization of LLaDA 8B using SING (still in progress)

Library Compatibility issues

- The GPUs in Sassauna servers are NVIDIA GeForce GTX 1080
- These GPUs are of cuda capability 6.1 and they result in errors when using cuda 12.8 or cuda 11.8 instead of using cu126
- The versions that worked with are pytorch 2.1.0 or pytorch 2.8.0 with cuda 12.6

Replicating SINQ Results on WikiText

Results are on Qwen3-1.7B

	Mem	Wiki2
3-bit	1.90	23.29
4-bit	2.04	17.19

		Qwen3-1.7B			Qwen3-14B			Qwen3-32B		
Method		Mem.	Wiki2 ↓	C4 ↓	Mem.	Wiki2 ↓	C4 ↓	Mem.	Wiki2 ↓	C4 ↓
Original (BF16)		3.44	16.67	19.21	29.54	8.64	12.01	65.52	7.60	10.77
3-BIT	RTN [†]	1.28	32.43	31.10	9.23	10.50	14.88	17.61	30.78	35.83
	Hadamard + RTN [†]	1.28	32.40	31.07	9.23	10.60	15.10	17.61	11.26	14.83
	HQQ	1.28	32.10	30.54	9.23	10.73	14.39	17.62	9.09	12.58
	SINQ (ours)	1.28	22.39	24.88	9.25	9.33	12.90	17.61	8.79	11.83
4-BIT	RTN [†]	1.42	18.74	20.81	10.54	8.95	12.50	20.78	8.92	12.80
	Hadamard + RTN [†]	1.42	19.10	20.70	10.54	8.85	12.35	20.78	8.28	11.60
	HQQ	1.42	18.96	22.10	10.54	8.78	12.36	20.78	8.62	12.20
	SINQ (ours)	1.42	17.14	19.83	10.56	8.76	12.21	20.73	7.74	10.96

†

Issues with Llada 8B loading

- When using transformers version 5.2.0 I get the following error:

“AttributeError: 'LLaDAModelLM' object has no attribute 'all_tied_weights_keys'.
Did you mean: '_tied_weights_keys'?”

- To solve this problem I use an older version of transformers version 4.46.2.
- Llada8B model size is 15GB which does not fit in a single GPU. I get OOM issues so I have to use 2 GPUs and set device_map=”auto”.

Inference Example on Llada8B

Prompts:

- "Lily can run 12 kilometers per hour for 4 hours. After that, she runs 6 kilometers per hour. How many kilometers can she run in 8 hours?"
- "Joy can read 8 pages of a book in 20 minutes. How many hours will it take her to read 120 pages?",
- "Randy has 60 mango trees on his farm. He also has 5 less than half as many coconut trees as mango trees. How many trees does Randy have in all on his farm?"

```
(sing) [msc26f13@sassauna3 SING]$ setenv CUDA_VISIBLE_DEVICES 1,3 ; python ./scripts/llada8B_eval.py
The model weights are not tied. Please use the `tie weights` method before using the `infer_auto_device` function.
loading checkpoint shards: 100%|██████████| 6/6 [00:04<00:00, 1.45it/s]
Lily can run 12 kilometers per hour for 4 hours, so she runs a total of  $12 * 4 = 48$  kilometers.
After that, she runs 6 kilometers per hour for the remaining 4 hours, so she runs a total of  $6 * 4 = 24$  kilometers.
Therefore, Lily can run a total of  $48 + 24 = 72$  kilometers in 8 hours.
The final result is 72
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If Joy can read 8 pages in 20 minutes, then she can read  $8/20 = 0.4$  pages in 1 minute.
To read 120 pages, it will take her  $120/0.4 = 300$  minutes.
To convert minutes to hours, we divide by 60, so 300 minutes is equal to  $300/60 = 5$  hours.
Therefore, it will take Joy 5 hours to read 120 pages.
Conclusively: 5
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Randy has 60 mango trees on his farm.
He also has 5 less than half as many coconut trees as mango trees.
Half as many coconut trees as mango trees is  $60/2 = 30$  coconut trees.
5 less than half as many coconut trees as mango trees is  $30 - 5 = 25$  coconut trees.
Therefore, Randy has 25 coconut trees on his farm.
In total, Randy has 60 mango trees + 25 coconut trees = 85 trees on his farm.
Summing up, we get 85
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(sing) [msc26f13@sassauna3 SING]$
```


Plan for next week

- Resolve issues related to quantization of LLaDA-8B models using SINQ
- Quantize Dream 7B model using SINQ
- Replicate results from Quantization Meets dLLMs paper and add the results of SINQ to the table
- Do exact replication of SINQ results if possible

Table 1: Model performance on general tasks under weight-only quantization.

Model	Setting	Method	WinoGrande	PIQA	ARC-C	ARC-E	Hellaswag	MMLU 5-shot	Avg	Drop
LLaDA-8B	FP Model	-	69.9	74.6	46.4	71.1	70.7	65.7	65.5	-
	W4A16 g128	GPTQ	69.7	73.9	47.9	72.5	70.4	64.7	65.3	↓ 0.3%
		AWQ	67.3	70.3	44.5	73.4	68.4	65.6	63.2	↓ 3.5%
	W3A16 g128	GPTQ	67.2	73.3	45.7	71.1	68.8	63.5	63.7	↓ 2.7%
		AWQ	66.4	69.2	42.8	71.8	66.4	64.0	61.8	↓ 5.6%
	FP Model	-	70.2	71.3	54.3	75.9	68.6	64.0	65.7	-
LLaDA-8B-Instruct	W4A16 g128	GPTQ	69.2	74.2	54.8	77.5	68.3	63.4	66.0	↑ 0.3%
		AWQ	68.8	71.0	53.3	76.1	68.1	63.4	64.9	↓ 0.1%
	W3A16 g128	GPTQ	67.4	73.7	50.7	76.4	66.7	62.1	64.1	↓ 2.4%
		AWQ	66.3	69.8	50.5	74.7	66.4	62.3	63.1	↓ 4.0%